

1 Notation

We extend the use of the for-all notation to work over ordered sequences like lists. This is usually used in conjunction with a list-comprehension-like notation to indicate the construction of sequence: e.g. $[\check{r} \mid \forall e \in [e]. \exists \check{r}. e : \check{r}]$; this denotes the construction of a list of types that corresponds to the list of expressions $[e]$.

2 Grammar

$\mathbb{S}$	<i>System Name</i>
S	<i>Subsystem Name</i>
X	<i>Explanation</i>
C	<i>Component Name</i>
I	<i>Indexed Semantic Properties</i>
A	<i>Semantic Property</i>
f	<i>Feature</i>
q	<i>Query</i>
c	<i>Command</i>
i	<i>Constraint</i>
p	<i>Semantic Property</i>
v	<i>Semantic Value</i>
a	<i>Semantic Assertion</i>
$\mathbb{S} \leftarrow X$	<i>Explanation Introduction 1</i>
$S \leftarrow X$	<i>Explanation Introduction 2</i>
$C \leftarrow X$	<i>Explanation Introduction 3</i>
$\mathbb{S} \leftarrow S$	<i>Subsystem Containment 1</i>
$S \leftarrow S$	<i>Subsystem Containment 2</i>
$S \leftarrow C$	<i>Component Containment</i>
$\mathbb{S} \leftarrow I$	<i>Indexing Introduction 1</i>
$S \leftarrow I$	<i>Indexing Introduction 2</i>
$I \leftarrow (p, v)$	<i>Semantic Property Introduction</i>
$C \leftarrow q$	<i>Query Introduction</i>
$C \leftarrow c$	<i>Command Introduction</i>
$C \leftarrow i$	<i>Constraint Introduction</i>
$S \leftarrow E$	<i>Event Set Introduction</i>
$E \leftarrow e$	<i>Event Introduction</i>
$S \leftarrow U$	<i>Scenarios Set Introduction</i>
$U \leftarrow u$	<i>Scenario Introduction</i>
$S \leftarrow R$	<i>Requirements Set Introduction</i>
$R \leftarrow r$	<i>Requirement Introduction</i>
$u ::= \{ e \}$	<i>Scenario as a List of Events</i>

$r ::= \{pp\} \rightarrow b$	<i>Requirement as a Predicate</i>
$pp ::= S \mid C$	<i>Parameters</i>
b	<i>Boolean Expression (Uninterpreted)</i>

The grammar captures entities from the Lando SSL grammar in terms of their names and describes their various relations. The names are expected to be unique. There are two exceptions: scenarios are captured as a list of events and requirements are captured as an arbitrary, uninterpreted predicate over a list of subsystems and components.

Further we required that a well-formed grammar have only a single System entity.

3 Type System

3.1 Type Grammar

$\tau ::= \hat{\tau} \mid \check{\tau} \mid \vec{\tau} \mid \dot{\tau}$	<i>Type</i>
$\hat{\tau} ::= \mathcal{B} \mid \mathcal{Z} \mid \mathcal{N} \mid System \mid Subsystem \mid Component \mid Event$	<i>Primitive Type</i>
$\vec{\tau} ::= [\tau]$	<i>List Type</i>
$\dot{\tau} ::= \vec{\tau} \rightarrow \mathcal{B}$	<i>Arrow Type</i>
$\check{\tau} ::= \check{\tau}_1, \check{\tau}_2, \dots$	<i>Derived Type</i>

3.2 Type Environment

In order to define the type system, we must first introduce a few auxilliary structures:

$CI ::= \{C \mapsto C\}$	<i>Component Inheritance Map</i>
$CT ::= \{C \mapsto \check{\tau}\}$	<i>Component Type Map</i>
$CD ::= \{\check{\tau} \mapsto \langle S, X, \{f\} \rangle\}$	<i>Component Data Map</i>
$f ::= q \mid c \mid i$	<i>Feature</i>
$SY ::= \langle \{S\}, X, I \rangle$	<i>System Data</i>
$ST ::= \{S \mapsto \check{\tau}\}$	<i>Subsystem Type Map</i>
$SD ::= \{\check{\tau} \mapsto \langle SP, X, \{S\}, \{C\}, \{E\}, \{U\}, \{R\}, I \rangle\}$	<i>Subsystem Data Map</i>
$SP ::= S \mid S$	<i>Parent of Subsystem</i>
$I ::= \{I \mapsto \langle SS, \{p \mapsto v\} \rangle\}$	<i>Indexing map</i>
$SS ::= S \mid S$	<i>Index Parent</i>
$EI ::= \langle EST, ESD, ET, ED \rangle$	<i>Event Info</i>
$EST ::= \{E \mapsto \check{\tau}\}$	<i>Event Set Type Map</i>

ESD	::=	$\{\check{\tau} \mapsto \langle S, \{e\} \rangle\}$	<i>Event Set Data Map</i>
ET	::=	$\{e \mapsto \check{\tau}\}$	<i>Event Type Map</i>
ED	::=	$\{\check{\tau} \mapsto E\}$	<i>Event Data Map</i>
UI	::=	$\langle UST, USD, UT, UD \rangle$	<i>Scenario Info</i>
UST	::=	$\{U \mapsto \check{\tau}\}$	<i>Scenario Set Type Map</i>
USD	::=	$\{\check{\tau} \mapsto \langle S, \{u\} \rangle\}$	<i>Scenario Set Data Map</i>
UT	::=	$\{u \mapsto \check{\tau}\}$	<i>Scenario Type Map</i>
UD	::=	$\{\check{\tau} \mapsto U\}$	<i>Scenario Data Map</i>
RI	::=	$\langle RST, RSD, RT, RD \rangle$	<i>Requirement Info</i>
RST	::=	$\{R \mapsto \check{\tau}\}$	<i>Requirement Set Type Map</i>
RSD	::=	$\{\check{\tau} \mapsto \langle S, \{r\} \rangle\}$	<i>Requirement Set Data Map</i>
RT	::=	$\{r \mapsto \check{\tau}\}$	<i>Requirement Type Map</i>
RD	::=	$\{\check{\tau} \mapsto R\}$	<i>Requirement Data Map</i>
Γ	::=	$\langle CI, CT, CD, T, SY, ST, SD, I, EI, UI, RI \rangle$	<i>Type Environment</i>

The type environment consists of a number of distinct elements required by our type system definition. For simplifying the rules, we define a shorthand notation for each of them.

Let $\Gamma = \langle CI, CT, CD, T, SY, ST, SD, I, EI, UI, RI \rangle$, then:

CI^Γ	$\doteq$	CI	
CT^Γ	$\doteq$	CT	
CD^Γ	$\doteq$	CD	
T^Γ	$\doteq$	T	
SY^Γ	$\doteq$	SY	
ST^Γ	$\doteq$	ST	
SD^Γ	$\doteq$	SD	
I^Γ	$\doteq$	I	
EST^Γ	$\doteq$	EST	where $EI = \langle EST, ESD, ET, ED \rangle$
ESD^Γ	$\doteq$	ESD	where $EI = \langle EST, ESD, ET, ED \rangle$
ET^Γ	$\doteq$	ET	where $EI = \langle EST, ESD, ET, ED \rangle$
ED^Γ	$\doteq$	ED	where $EI = \langle EST, ESD, ET, ED \rangle$
UST^Γ	$\doteq$	UST	where $UI = \langle UST, USD, UT, UD \rangle$
USD^Γ	$\doteq$	USD	where $UI = \langle UST, USD, UT, UD \rangle$
UT^Γ	$\doteq$	UT	where $UI = \langle UST, USD, UT, UD \rangle$
UD^Γ	$\doteq$	UD	where $UI = \langle UST, USD, UT, UD \rangle$
RST^Γ	$\doteq$	RST	where $RI = \langle RST, RSD, RT, RD \rangle$
RSD^Γ	$\doteq$	RSD	where $RI = \langle RST, RSD, RT, RD \rangle$
RT^Γ	$\doteq$	RT	where $RI = \langle RST, RSD, RT, RD \rangle$
RD^Γ	$\doteq$	RD	where $RI = \langle RST, RSD, RT, RD \rangle$

3.3 Rules

Currently all rules have the form: $e : \tau$. This is not strictly necessary; for example many rules can be relations that does not necessarily involve types at all (e.g. Intro explanation); in such cases the types returned are somewhat artificial. In the long run, we can elide types for such rules as long as we can ensure that the set of rules are inductive in the right way. But since this is still the first phase of the type system, I am keeping the rules consistent and always including a type for the time being.

$$\begin{array}{c} \text{SYSTEM INTRODUCTION} \\ \hline \Gamma \vdash \mathbb{S} : \text{System} \end{array} \quad \begin{array}{c} \text{SUBSYSTEM INTRODUCTION} \\ \mathbb{S} \mapsto \check{\tau} \in \mathbb{ST}^\Gamma \quad \check{\tau} <: \text{Subsystem} \in \mathbb{T} \\ \hline \Gamma \vdash \mathbb{S} : \check{\tau} \end{array}$$

$$\begin{array}{c} \text{EXPLANATION INTRODUCTION 1} \\ \mathbb{S} \mapsto \langle _, X, _ \rangle \in \mathbb{SY}^\Gamma \quad X = X' \\ \hline \Gamma \vdash \mathbb{S} \leftarrow X' : \text{System} \end{array}$$

$$\begin{array}{c} \text{EXPLANATION INTRODUCTION 2} \\ \mathbb{S} \mapsto \check{\tau} \in \mathbb{ST}^\Gamma \\ \check{\tau} \mapsto \langle \text{SP}, X, \{\mathbb{S}\}, \{\mathbb{C}\}, \{\mathbb{E}\}, \{\mathbb{U}\}, \{\mathbb{R}\}, \mathbb{I} \rangle \in \mathbb{SD}^\Gamma \quad X = X' \\ \hline \Gamma \vdash \mathbb{S} \leftarrow X' : \check{\tau} \end{array}$$

$$\begin{array}{c} \text{SUBSYSTEM CONTAINMENT 1} \\ \mathbb{SY}^\Gamma = \langle \{\mathbb{S}\}, X, \mathbb{I} \rangle \quad \mathbb{S} \in \{\mathbb{S}\} \\ \mathbb{S} \mapsto \check{\tau} \in \mathbb{ST}^\Gamma \quad \check{\tau} \mapsto \langle \text{SP}, _, _, _, _, _, _ \rangle \in \mathbb{SD}^\Gamma \quad \text{SP} = \mathbb{S} \\ \hline \mathbb{S} \leftarrow \mathbb{S} : \text{System} \end{array}$$

$$\begin{array}{c} \text{SUBSYSTEM CONTAINMENT 2} \\ \mathbb{S} \mapsto \check{\tau} \in \mathbb{ST}^\Gamma \\ \check{\tau} \mapsto \langle \text{SP}, X, \{\mathbb{S}\}, \{\mathbb{C}\}, \{\mathbb{E}\}, \{\mathbb{U}\}, \{\mathbb{R}\}, \mathbb{I} \rangle \in \mathbb{SD}^\Gamma \quad \mathbb{S}' \in \{\mathbb{S}\} \\ \mathbb{S}' \mapsto \check{\tau}' \in \mathbb{ST}^\Gamma \quad \check{\tau}' \mapsto \langle \text{SP}, _, _, _, _, _, _ \rangle \in \mathbb{SD}^\Gamma \quad \text{SP} = \mathbb{S} \\ \hline \Gamma \vdash \mathbb{S} \leftarrow \mathbb{S}' : \check{\tau} \end{array}$$

$$\begin{array}{c} \text{EXPLANATION INTRODUCTION 3} \\ \mathbb{C} \mapsto \check{\tau} \in \mathbb{CT}^\Gamma \quad \check{\tau} \mapsto \langle \mathbb{S}, X, \{\mathbb{f}\} \rangle \in \mathbb{CD}^\Gamma \quad X = X' \\ \hline \Gamma \vdash \mathbb{C} \leftarrow X' : \check{\tau} \end{array}$$

$$\begin{array}{c} \text{COMPONENT SUBTYPING} \\ \mathbb{C} \mapsto \check{\tau} \in \mathbb{CT}^\Gamma \\ \mathbb{C} \mapsto \mathbb{C}' \in \mathbb{CI}^\Gamma \quad \mathbb{C}' \mapsto \check{\tau}' \in \mathbb{CT}^\Gamma \quad \text{NCI}(\mathbb{CI}^\Gamma, \mathbb{C}, \mathbb{C}') \quad \check{\tau} <: \check{\tau}' \in \mathbb{T}^\Gamma \\ \check{\tau} \mapsto \langle _, _, \{\mathbb{f}\} \rangle \in \mathbb{CD}^\Gamma \quad \check{\tau}' \mapsto \langle _, _, \{\mathbb{f}'\} \rangle \in \mathbb{CD}^\Gamma \quad \{\mathbb{f}'\} \subset \{\mathbb{f}\} \\ \hline \Gamma \vdash \mathbb{C} : \check{\tau} \end{array}$$

COMPONENT IMPLICIT SUBTYPING

$$\frac{C \mapsto \check{\tau} \in \mathbb{CT}^\Gamma \quad \nexists C'. C \mapsto C' \in \mathbb{CI}^\Gamma \quad \check{\tau} <: \text{Component} \in \mathbb{T}^\Gamma}{\Gamma \vdash C : \check{\tau}}$$

COMPONENT CONTAINMENT

$$\frac{\begin{array}{l} S \mapsto \check{\tau} \in \mathbb{ST}^\Gamma \quad \check{\tau} \mapsto \langle \mathbf{SP}, \mathbf{X}, \{\mathbf{S}\}, \{\mathbf{C}\}, \{\mathbf{E}\}, \{\mathbf{U}\}, \{\mathbf{R}\}, \mathbf{I} \rangle \in \mathbb{SD}^\Gamma \\ C \in \{\mathbf{C}\} \quad C \mapsto \check{\tau}' \in \mathbb{CT}^\Gamma \quad \check{\tau}' \mapsto \langle \mathbf{S}', _, _ \rangle \in \mathbb{CD}^\Gamma \quad \mathbf{S} = \mathbf{S}' \end{array}}{\Gamma \vdash S \leftarrow C : \check{\tau}'}$$

QUERY INTRODUCTION

$$\frac{C \mapsto \check{\tau} \in \mathbb{CT}^\Gamma \quad \check{\tau} \mapsto \langle \mathbf{S}, \mathbf{X}, \{\mathbf{f}\} \rangle \in \mathbb{CD}^\Gamma \quad \mathbf{q} \in \{\mathbf{f}\}}{\Gamma \vdash C \leftarrow \mathbf{q} : \check{\tau}}$$

COMMAND INTRODUCTION

$$\frac{C \mapsto \check{\tau} \in \mathbb{CT}^\Gamma \quad \check{\tau} \mapsto \langle \mathbf{S}, \mathbf{X}, \{\mathbf{f}\} \rangle \in \mathbb{CD}^\Gamma \quad \mathbf{c} \in \{\mathbf{f}\}}{\Gamma \vdash C \leftarrow \mathbf{c} : \check{\tau}}$$

CONSTRAINT INTRODUCTION

$$\frac{C \mapsto \check{\tau} \in \mathbb{CT}^\Gamma \quad \check{\tau} \mapsto \langle \mathbf{S}, \mathbf{X}, \{\mathbf{f}\} \rangle \in \mathbb{CD}^\Gamma \quad \mathbf{i} \in \{\mathbf{f}\}}{\Gamma \vdash C \leftarrow \mathbf{i} : \check{\tau}}$$

INDEXING INTRODUCTION 1

$$\frac{S \mapsto \langle \{\mathbf{S}\}, \mathbf{X}, \mathbf{I} \rangle \in \mathbb{SY}^\Gamma \quad \mathbf{I} = \mathbf{I}' \quad \mathbf{I} \mapsto \langle \mathbf{SS}, _ \rangle \in \mathbb{I}^\Gamma \quad \mathbf{SS} = \mathbf{S}}{\Gamma \vdash S \leftarrow \mathbf{I}' : \text{System}}$$

INDEXING INTRODUCTION 2

$$\frac{\begin{array}{l} S \mapsto \check{\tau} \in \mathbb{ST}^\Gamma \quad \check{\tau} \mapsto \langle \mathbf{SP}, \mathbf{X}, \{\mathbf{S}\}, \{\mathbf{C}\}, \{\mathbf{E}\}, \{\mathbf{U}\}, \{\mathbf{R}\}, \mathbf{I} \rangle \in \mathbb{SD}^\Gamma \\ \mathbf{I} = \mathbf{I}' \quad \mathbf{I} \mapsto \langle \mathbf{SS}, _ \rangle \in \mathbb{I}^\Gamma \quad \mathbf{SS} = \mathbf{S} \end{array}}{\Gamma \vdash S \leftarrow \mathbf{I}' : \check{\tau}}$$

SEMANTIC PROPERTY INTRODUCTION

$$\frac{I \mapsto \langle \mathbf{SS}, \{\mathbf{p} \mapsto \mathbf{v}\} \rangle \in \mathbb{I}^\Gamma \quad \mathbf{p} \mapsto \mathbf{v} \in \{\mathbf{p} \mapsto \mathbf{v}\} \quad \check{\tau} \mapsto \text{TypeOf}(\Gamma, \mathbf{SS})}{\Gamma \vdash I \leftarrow (\mathbf{p}, \mathbf{v}) : \check{\tau}}$$

EVENT SET INTRODUCTION

$$\frac{\begin{array}{l} S \mapsto \check{\tau} \in \mathbb{ST}^\Gamma \quad \check{\tau} \mapsto \langle \mathbf{SP}, \mathbf{X}, \{\mathbf{S}\}, \{\mathbf{C}\}, \{\mathbf{E}\}, \{\mathbf{U}\}, \{\mathbf{R}\}, \mathbf{I} \rangle \in \mathbb{SD}^\Gamma \\ E \in \{\mathbf{E}\} \quad E \mapsto \check{\tau}' \in \mathbb{EST}^\Gamma \quad \check{\tau}' \mapsto \langle \mathbf{S}', _ \rangle \in \mathbb{ESD}^\Gamma \quad \mathbf{S} = \mathbf{S}' \end{array}}{\Gamma \vdash S \leftarrow E : \check{\tau}'}$$

EVENT INTRODUCTION

$$\frac{\begin{array}{l} E \mapsto \check{\tau} \in \mathbb{EST}^\Gamma \quad \check{\tau} \mapsto \langle \mathbf{S}, \{\mathbf{e}\} \rangle \in \mathbb{ESD}^\Gamma \quad \mathbf{e} \in \{\mathbf{e}\} \\ \mathbf{e} \mapsto \check{\tau}' \in \mathbb{ET}^\Gamma \quad \check{\tau}' \mapsto E' \in \mathbb{ED}^\Gamma \quad E = E' \quad \check{\tau}' <: \text{Event} \in \mathbb{T}^\Gamma \end{array}}{\Gamma \vdash E \leftarrow \mathbf{e} : \check{\tau}'}$$

SCENARIO SET INTRODUCTION

$$\frac{\begin{array}{l} S \mapsto \check{\tau} \in \mathbb{ST}^\Gamma \quad \check{\tau} \mapsto \langle \mathbf{SP}, \mathbf{X}, \{\mathbf{S}\}, \{\mathbf{C}\}, \{\mathbf{E}\}, \{\mathbf{U}\}, \{\mathbf{R}\}, \mathbf{I} \rangle \in \mathbb{SD}^\Gamma \\ U \in \{\mathbf{U}\} \quad U \mapsto \check{\tau}' \in \mathbb{UT}^\Gamma \quad \check{\tau}' \mapsto \langle \mathbf{S}', _ \rangle \in \mathbb{USD}^\Gamma \quad S = S' \end{array}}{\Gamma \vdash S \leftarrow U : \check{\tau}}$$

SCENARIO INTRODUCTION

$$\frac{\begin{array}{l} U \mapsto \check{\tau} \in \mathbb{UT}^\Gamma \quad \check{\tau} \mapsto \langle \mathbf{S}, \{\mathbf{u}\} \rangle \in \mathbb{USD}^\Gamma \\ \mathbf{u} \in \{\mathbf{u}\} \quad \mathbf{u} = \{\mathbf{e}\} \quad \check{\tau} = [\check{\tau} \mid \forall \mathbf{e} \in \{\mathbf{e}\}. \mathbf{e} \mapsto \check{\tau} \in \mathbb{ET}^\Gamma] \\ \mathbf{u} \mapsto \check{\tau} \in \mathbb{UT}^\Gamma \quad \check{\tau} \mapsto \mathbf{U}' \in \mathbb{UD}^\Gamma \quad U = U' \end{array}}{\Gamma \vdash U \leftarrow \mathbf{u} : \check{\tau}}$$

REQUIREMENT SET INTRODUCTION

$$\frac{\begin{array}{l} S \mapsto \check{\tau} \in \mathbb{ST}^\Gamma \quad \check{\tau} \mapsto \langle \mathbf{SP}, \mathbf{X}, \{\mathbf{S}\}, \{\mathbf{C}\}, \{\mathbf{E}\}, \{\mathbf{U}\}, \{\mathbf{R}\}, \mathbf{I} \rangle \in \mathbb{SD}^\Gamma \\ R \in \{\mathbf{R}\} \quad R \mapsto \check{\tau}' \in \mathbb{RT}^\Gamma \quad \check{\tau}' \mapsto \langle \mathbf{S}, _ \rangle \in \mathbb{RSD}^\Gamma \quad S = S' \end{array}}{\Gamma \vdash S \leftarrow R : \check{\tau}}$$

REQUIREMENT INTRODUCTION

$$\frac{\begin{array}{l} R \mapsto \check{\tau} \in \mathbb{RT}^\Gamma \quad \check{\tau} \mapsto \langle \mathbf{S}, \{\mathbf{r}\} \rangle \in \mathbb{RSD}^\Gamma \\ \mathbf{r} \in \{\mathbf{r}\} \quad \check{\tau} = [\check{\tau} \mid \forall \mathbf{pp} \in \{\mathbf{pp}\}. \text{TYPEOF}(\mathbf{pp}) = \check{\tau}] \\ \mathbf{b} : \mathcal{B} \quad \dot{\tau} = \check{\tau} \rightarrow \mathcal{B} \quad \mathbf{r} \mapsto \dot{\tau} \in \mathbb{RT}^\Gamma \quad \dot{\tau} \mapsto \mathbf{R}' \in \mathbb{RD}^\Gamma \quad R = R' \end{array}}{\Gamma \vdash R \leftarrow \mathbf{r} : \check{\tau}}$$

3.4 Auxiliary Functions and Relations

Non-circular-Inheritance indicates that a component's inheritance chain does not have a cycle. Note that the particular relation only checks for circles that includes the specific component. However since this relation must be satisfied by every component, any cycle in the chain will be detected at some point. Formally non-circularity is a ternary relation defined inductively as below:

$$\text{NCI}(\mathbb{CI}^\Gamma, C, C') = \begin{cases} \text{undefined} & \text{iff } C = C' \\ \text{undefined} & \text{iff } \text{NCI}(\mathbb{CI}^\Gamma, C, C'') \text{ and } C' \mapsto C'' \in \mathbb{CI}^\Gamma \\ \text{defined} & \text{otherwise} \end{cases}$$

TYPEOF is a simple function to extract the type associated with an entity in the environment

$$\begin{array}{ll} \text{TYPEOF}(\Gamma, S) = \text{System} \\ \text{TYPEOF}(\Gamma, S) = \check{\tau} & \text{iff } S \mapsto \check{\tau} \in \mathbb{ST}^\Gamma \\ \text{TYPEOF}(\Gamma, C) = \check{\tau} & \text{iff } S \mapsto \check{\tau} \in \mathbb{CT}^\Gamma \end{array}$$